

Apple][/][+ — Pinouts

6502 · 4116 RAM · 2316/2716 ROM · MM5740 KEYBOARD

DIP pinouts from the top, notch up, pin 1 top-left. Power **red**, ground grey, clocks blue. Verify against the datasheet before powering up.

Legend — reading the pinouts notation

/NAME Active LOW — true / asserted when the pin is at logic **LOW** (~0 V). A leading / (datasheets often use an overline) marks these: e.g. /RES /IRQ /CS /WE.

NAME Active HIGH / normal — asserted when the pin is **HIGH** (~+5 V). No slash. R/W is a level: HIGH = read, LOW = write.

Colour key: **Vcc / Vdd = power** · **GND / Vss = ground** · **φ = clock**. **Orientation:** chip seen from the top, notch up — pin 1 top-left, numbers run **down the left** then **up the right**.

6502 CPU

40-DIP

Vss	1	40	/RES
RDY	2	39	φ2 out
φ1 out	3	38	S0
/IRQ	4	37	φ0 in
NC	5	36	NC
/NMI	6	35	NC
SYNC	7	34	R/W
Vcc	8	33	D0
A0	9	32	D1
A1	10	31	D2
A2	11	30	D3
A3	12	29	D4
A4	13	28	D5
A5	14	27	D6
A6	15	26	D7
A7	16	25	A15
A8	17	24	A14
A9	18	23	A13
A10	19	22	A12
A11	20	21	Vss

Apple][/][+ CPU @ 1.02 MHz. Everything else is discrete TTL + PROMs (no custom video chip). **Fails:** dead machine, no boot beep.

4116 DRAM 16K×1

16-DIP

Vbb -5V	1	16	Vss
DIN	2	15	/CAS
/WE	3	14	DOUT
/RAS	4	13	A6
A0	5	12	A3
A2	6	11	A4
A1	7	10	A5
Vdd +12V	8	9	Vcc +5V

48KB main RAM = three banks of eight 4116 (needs +5/-5/+12V). RAM config jumpers set the size. **Fails:** won't boot / garbage / bad memory test — swap 4116s.

2716/2316 ROM 2K×8

24-DIP

A7	1	24	Vcc
A6	2	23	A8
A5	3	22	A9
A4	4	21	Vpp
A3	5	20	/OE
A2	6	19	A10
A1	7	18	/CE
A0	8	17	07
00	9	16	06
01	10	15	05
02	11	14	04
Vss	12	13	03

Monitor + Applesoft in 2316B mask ROMs (2K each, \$D000-\$F800), shown as the compatible 2716 pinout. **2316B chip-selects differ on pins 18/20/21 — verify.**

MM5740 keyboard enc.

40-DIP

B3	1	40	B4
VLL	2	39	B9
CLOCK	3	38	B2
X9	4	37	B1
X8	5	36	B8
X7	6	35	B7
X6	7	34	B6
X5	8	33	B5
X4	9	32	VSS
X3	10	31	Y9
X2	11	30	Y8
X1	12	29	Y7
DStrb0	13	28	Y6
DStrbC	14	27	Y5
OutEn	15	26	Y4
Repeat	16	25	Y3
KeyBnce	17	24	Y2
VGG	18	23	Y1
Control	19	22	Y10
ShiftLk	20	21	SHIFT

Keyboard encoder on early][/][+ Datamatics boards (9-bit code, B1-B9). PMOS: VLL(2)/VGG(18)/VSS(32). The later //e//c use the AY-5-3600 instead (see the //e sheet). **Fails:** dead/stuck keyboard.

Memory & I/O

map

\$0000-BFFF 48KB RAM (4116)

\$C000-CFFF I/O + slot space

\$D000-F7FF Applesoft/Monitor ROM

Slots 8× 50-pin peripheral connectors

Video discrete TTL + 2513 char-gen ROM

No MMU/IOU on the][/][+ — those are //e custom chips (see the Apple //e sheet).

Test & Repair

tips

No beep / dead 6502 / clock / ROM

Garbage / no boot 4116 RAM (swap banks)

No video discrete video TTL / char-gen ROM

Dead keys keyboard encoder

Apple][faults are usually RAM, ROM, the 6502, or a 74-series TTL — there is no single video chip.

Notes

△ verify these

2316B ROM CS pins 18/20/21 vs 2716

Keyboard enc. MM5740 (early) vs AY-5-3600 (//e)

Amber pins and △ **flagged chips** are my best reconstruction from incomplete/garbled sources — confirm each against a datasheet before use.